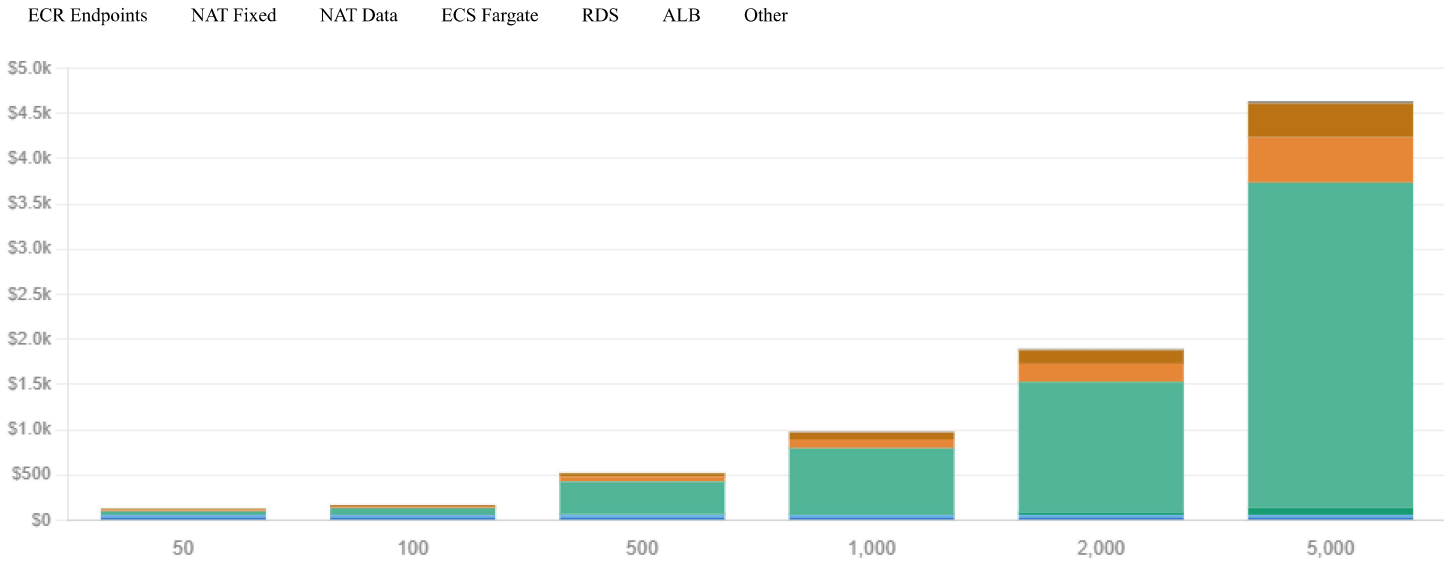
COST ANALYSIS

Hybrid VPC Architecture — Coding Assessment Platform

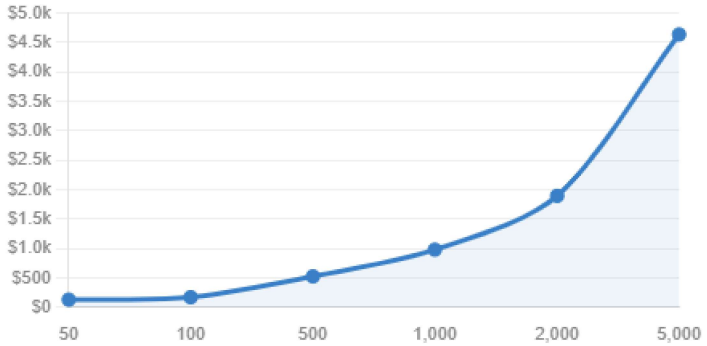
Monthly cost projections for 50–5,000 active users in us-east-1, using NAT gateway + ECR VPC endpoints. All prices on-demand, 24/7 operation.

50 users / month	1,000 users / month	5,000 users / month	Fixed baseline cost
$130	$980	$4,637	$78
$2.60 per user	$0.98 per user	$0.93 per user	VPC + NAT + ALB base

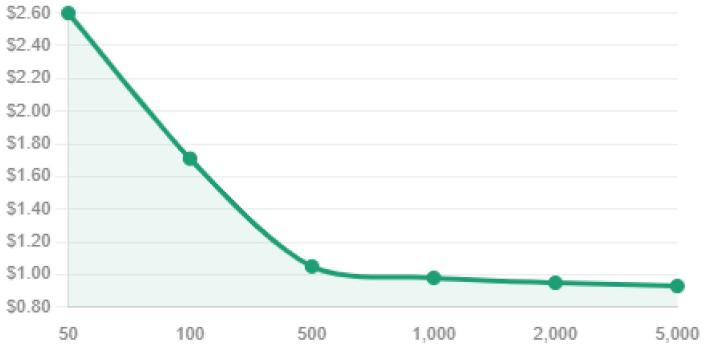
COST BREAKDOWN BY USER SCALE



TOTAL MONTHLY COST CURVE



AVERAGE COST PER USER



DETAILED COST TABLE

USERS	ECR ENDPOINTS	NAT FIXED	NAT DATA	ECS FARGATE	RDS	ALB	OTHER	TOTAL
50	$29	$33	$1	$36	$12	$17	$2	$130
100	$29	$33	$2	$72	$12	$20	$3	$171
500	$29	$33	$8	$360	$50	$40	$5	$525
1,000	$29	$33	$15	$720	$100	$75	$8	$980
2,000	$29	$33	$30	$1,440	$200	$150	$12	$1,894
5,000	$29	$33	$75	$3,600	$500	$375	$25	$4,637

ARCHITECTURE COMPARISON — 1,000 USERS

ARCHITECTURE	MONTHLY COST	PROS	CONS
Hybrid — NAT + ECR endpoints recommended	$980	Balanced cost; avoids large variable charges; simple to manage	NAT gateway is single point of failure
Full endpoints (no NAT)	$1,050	Fully private; no data-processing charges	Higher fixed cost (~$105/month for endpoints)
NAT only (no endpoints)	$1,050–1,200	Lower fixed cost; simpler networking	Variable data charges can spike; image pulls incur processing fees

KEY INSIGHTS

ECS Fargate dominates costs

Compute accounts for 28–78% of total spend across all scales. One task per 50 users at $36/month each is the primary scaling lever. Graviton (arm64) instances can cut compute 10–20%.

Fixed costs are modest

VPC endpoints ($29), NAT gateway ($33), and ALB base ($16) total $78/month regardless of user count. This baseline is well-amortized beyond ~100 users.

Per-user cost falls sharply

Average cost per user drops from $2.60 at 50 users to $0.93 at 5,000 users — a 64% reduction — as fixed infrastructure costs spread over a larger base.

NAT data costs stay contained

By routing ECR image pulls through VPC endpoints, NAT data charges remain small ($1–$75/month). Only lightweight traffic (Secrets Manager, CloudWatch) passes through the NAT.